2026 Quantum Mechanics

Welcome to the **British Physics Olympiad Computational Challenge 2026.** The goal is to build *computer models* based upon the instructions in the **Challenge Presentation document.** These can mostly be achieved using a *spreadsheet* such as Microsoft Excel, although you are very much encouraged to use a *programming language* of your choice*.

The challenge runs from **Easter 2026 till August 2026.** To submit an entry you will need to fill in a web form at **https://www.bpho.org.uk/.**

Additional resources can be found at: **https://eclecticon.info/physics_bpho_compphys.htm**

The deliverable of the challenge is to produce a **screencast** of ***maximum length three minutes*** which describes your response to the challenge, i.e. the graphs and the code & spreadsheets and your explanation of these. Your video should make it really clear *how you* have arrived at your solutions to the tasks set. This is what we need evidence for in your video. All credit is for 'show your working' !
No videos over three minutes can get a Gold. Deliberately speeded up videos will be disqualified.

The videos must be uploaded to **YouTube**, and we recommend you set these as *Unlisted* with *Comments disabled*. **Your entry will comprise a YouTube link.** To produce the screencast, we recommend the Google Chrome add-on **Screencastify.**

You can enter the challenge **individually** or in **pairs**. If you opt for the latter, *both* of you must make equal contributions to the screencast.

Gold, **Silver** or **Bronze** e-certificates will be emailed to each complete entry, and the **top five** Golds will be invited to present their work at a special ceremony. Note no additional feedback will be provided, and the decision of the judges is final.

Bronze: All spreadsheet-based challenge elements completed. Some basic coding attempted.

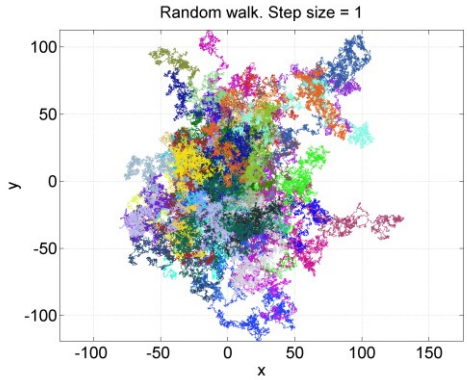
Silver: All tasks completed in code.

Gold: All tasks completed to a high standard, with possible extension work such as the construction of apps (i.e. programs with graphical user interfaces) and significant development of the models. The highest quality entries will typically contain research papers based upon the models and computational methods. For this challenge we also provide several Quantum Mechanics problem sheets, which could be attempted and written up. Many of the problems link directly to the challenges (e.g. extension to Task#7 relating to verifying the Uncertainty Principle for a 'particle in a box.').

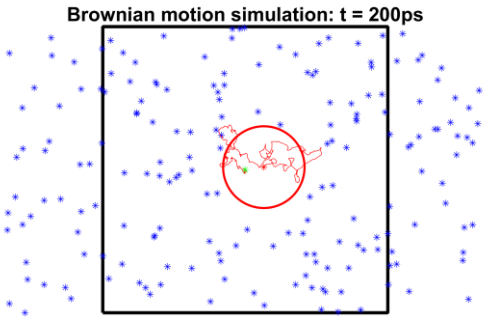
*MATLAB or Python is recommended, although any system that can easily execute code in loops and plot graphs will do. e.g. Octave, Java, Javascript, C#, C++, Mathematica... Use what you can access and feel comfortable with. These Programming resources might be a helpful start.

INSTRUCTIONS *** First download the Challenge Presentation from the BPhO website ***

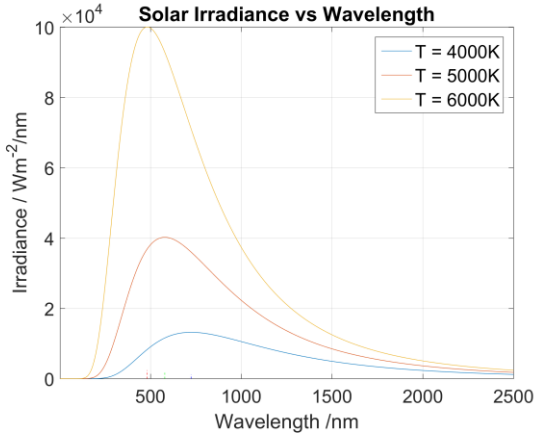
Summary of tasks (each will have Bronze, Silver and Gold aspects - although each task is more involved than the previous). All mathematical details for tasks are provided in the Challenge Presentation pack.



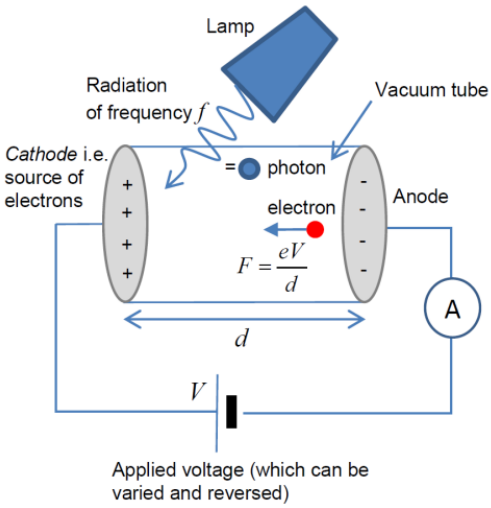
TASK #1. Random walk. Create a model of a random walk of N steps of size s. Each step is in a random direction, with angle θ from the horizontal chosen from a uniform distribution between 0 and 2π radians.



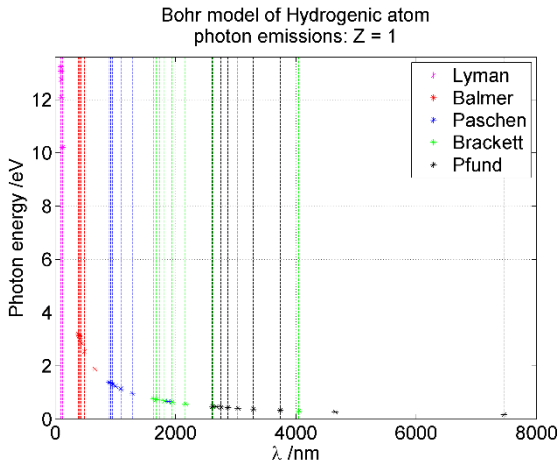
TASK #2. Brownian motion. Consider N small particles of mass m and radius r moving randomly, some of which collide with a large particle of mass M and radius R. Use a random walk to model the smaller particles, and conservation of momentum to model collisions with the larger particle. Animate the subsequent motion of the system.



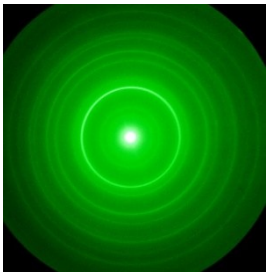
TASK #3. Plot the **Planck 'Black Body radiation' spectrum** $B(\lambda,T)$ for several temperatures, and also plot Einstein's model of molar heat capacity of solids vs temperature for various atomic crystals such as gold, copper, iron.



TASK #4. Photoelectric effect. Plot photoelectron stopping voltage vs frequency (or in-vacuum wavelength) of incident photons for various metals. **EXTENSION:** Create an animated app like the PhET demo.

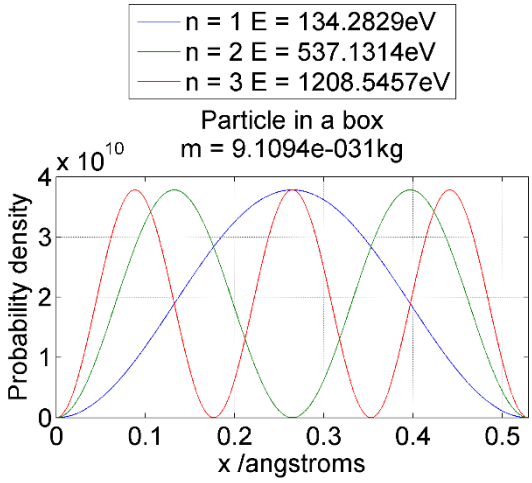


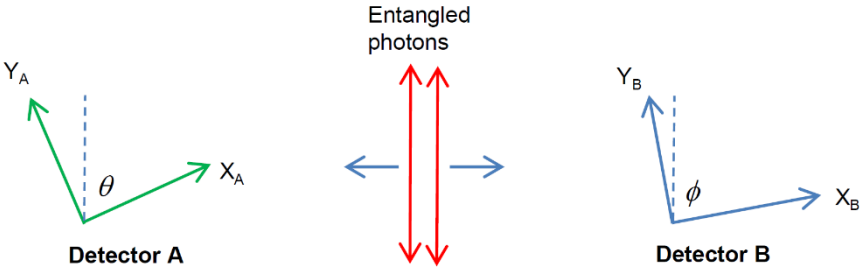
TASK #5. Hydrogen emission spectrum and Bohr's model of a hydrogenic atom. Create a graph of photon energy vs wavelength for photon emissions from hydrogen atoms due to transitions between electron energy levels.



TASK #6. Electron diffraction. Create a computer model of the 'electron wave' rings on a phosphor screen with accelerating voltage V as a variable. (1kV to 5kV). Assume atomic spacings d are layer spacings of graphite e.g. 0.123nm or 0.213nm. Check your model by generating a $1/\sqrt{V}$ vs $\sin\frac{1}{2}\phi$ straight line graph, where ring radius is $x = r\sin 2\phi$ and spherical glass diffraction tube radius is $r = 65$ mm.

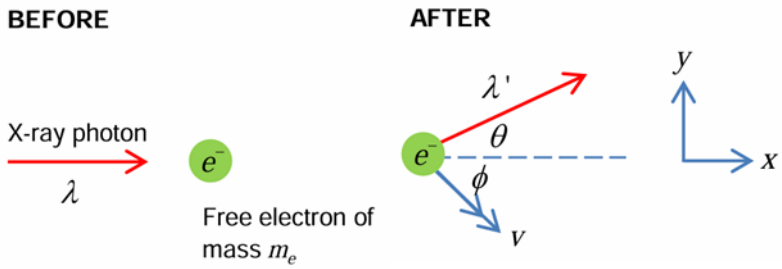
TASK #7. Particle-in-a-box solution to the Schrödinger Equation. Plot energy vs quantum number, and probability densities $|\psi|^2$ vs displacement x in the box. **EXTENSION**: Write up the model, and show that a particle in the box obeys the *Uncertainty Principle* i.e. $\Delta x \Delta p \geq \frac{1}{2}\hbar$, where Δp is the momentum uncertainty.

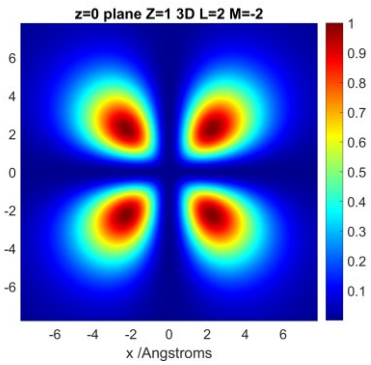




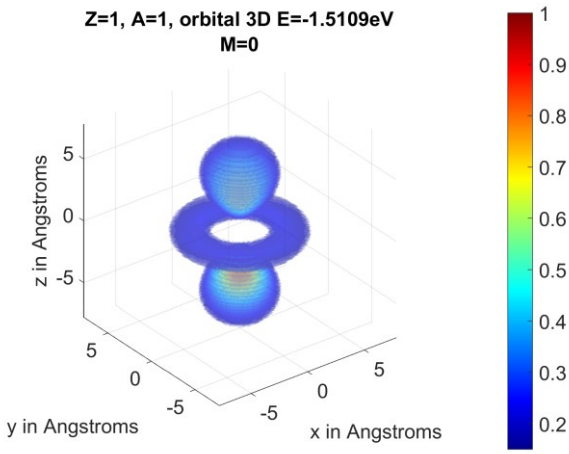
TASK #8. Quantum cryptography. Create a visual calculator of the classical and quantum mismatch probabilities for the detection of polarized entangled photons.

TASK #9. Compton scattering. Plot fractional wavelength shift $\Delta\lambda/\lambda$, electron recoil speed v and electron recoil angle ϕ vs photon scattering angle θ.





TASK #10. Hydrogenic orbitals. Plot 2D slices and 3D visualizations of the probability density $|\psi|^2$ for an electron in a hydrogenic atom, given n, l, m *quantum numbers*. Use the provided solutions to the Schrödinger Equation as a recipe (i.e. no need to derive).



POSSIBLE EXTENSION OPPORTUNITIES:

- Answer the questions in the **Quantum Mechanics problem sheets** (provided with this document) and write these up in an illustrated paper. Many link to the challenge tasks, so you can re-use your models. A good opportunity to learn LaTeX – which is the typesetting language used to write most technical papers and books in the physical sciences. Including Science by Simulation.

- Write a graphical user interface (GUI) for some of the Quantum models and code these as an 'app'. Creating an iOS/Android smartphone app or interactive website will particularly impress the judges.

- Investigate combinations of hydrogenic orbitals (see TASK #10) that approximate multi-electron atoms, and indeed multi-atom molecules. (Note this could be a HUGE project – so only very initial progress is expected). Or animate the morphing between different m numbers for given l and n quantum numbers. For inspiration, see *Orbitals* by C.S. McCaw.